

ITA0506-COMPUTER VISION LAB EXPERIMENT

DATE :15-07-2026

EXPNO:5

AIM: Perform basic Image Handling and processing operations on the image

- **Read an image in python and Erode an Image using erode function.**

Algorithm:

- **Start and import libraries**
Start the program and import the required modules: cv2 for image processing and numpy for creating the structuring element (kernel).
- **Read the input image**
Read the image from the specified file path using cv2.imread() and store it in a variable.
- **Check if image is loaded**
Verify that the image is successfully loaded (not None). If loading fails, print an error message and terminate the program.
- **Convert image to grayscale**
Convert the color image from BGR format to grayscale using cv2.cvtColor(img, cv2.COLOR_BGR2GRAY) to simplify further processing.
- **Define kernel and apply dilation**
Create a structuring element (for example, a 3×3 matrix of ones) using numpy, then apply the dilation operation on the grayscale image using cv2.dilate(gray, kernel, iterations=1) to thicken bright regions.
- **Display results and exit**
Display the original, grayscale, and dilated images using cv2.imshow(), wait for a key press using cv2.waitKey(0), and finally close all display windows using cv2.destroyAllWindows() to end the program.

Code:

```

import cv2

import numpy as np

img = cv2.imread(r"C:\Users\surya\OneDrive\Pictures\Screenshots
1\Screenshot 2026-07-23 114834.png")

if img is None:

    print("Error: Image not found or unable to open.")
    exit()

gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

kernel = np.ones((3, 3), np.uint8)

eroded = cv2.erode(gray, kernel, iterations=1)

cv2.imshow("Original Image", img)

cv2.imshow("Grayscale Image", gray)

cv2.imshow("Eroded Image", eroded)

cv2.waitKey(0)

cv2.destroyAllWindows()

```

OUTPUT :

